

Ruibin Yuan

PhD Student in Artificial Intelligence, HKUST | AI Researcher, Developer, and Musician

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🌐 github.com/a43992899 | 🔍 Google Scholar | 🧑‍🔬 MAP Research Community 📞 +852 46830974 | +86 18033231161

Research focus. Foundation models for music generation and understanding; multimodal LLMs; large-scale data, training infrastructure, and open research communities for creative AI.

Education

Hong Kong University of Science and Technology (HKUST) <i>PhD in Artificial Intelligence</i> Supervisors: Jie Fu, Wei Xue, Yike Guo	09/2023–Present Hong Kong
Carnegie Mellon University (CMU) <i>M.S. in Music and Technology, Computer Science Emphasis</i> Supervisors: Roger B. Dannenberg, Gus Xia, Riccardo Schulz	09/2021–05/2023 Pittsburgh, PA
Beijing University of Posts and Telecommunications (BUPT) <i>B.S. in Electronic Science and Technology</i>	09/2016–06/2020 Beijing, China

Academic Experience

Qwen Music <i>Team Lead.</i> Lead foundation model research for music generation and contribute music understanding capabilities in the Qwen-Omni series.	04/2025–Present
Multimodal Art Projection Research Community (MAP) 🌐 <i>Cofounder.</i> Build open large language, music, and multimodal models, with a focus on open-source training, data collection, and creative AI. Lead MAP’s multimodal and AI music research direction across projects including COIG, MERT, ChatMusician, CMMMU, and YuE.	09/2022–Present
YuE (OpenSuno) 🌐 <i>Team Lead.</i> Built an open full-song generation foundation model for lyrics-to-song and long-form music generation, attracting broad open-source adoption.	03/2024–03/2025
NEO LLM 🌐 <i>Core Member.</i> Helped train a fully open LLM from scratch, contributing to infrastructure, PB-scale data collection and cleaning, training, evaluation, experiment logging, and tooling. Operated large GPU clusters and monitored hardware/software failures during training.	09/2023–05/2024
ChatMusician 🌐 <i>Team Lead.</i> Trained the first symbolic music LLM; collected a large-scale symbolic music and language dataset and an advanced music understanding benchmark. The project received public attention from Yann LeCun and AK.	2023–2024
MMMU & CMMMU 🌐 Contributed to multimodal benchmark development extending MMLU/CMMLU-style expert reasoning to multimodal settings. MMMU was recommended by Jeff Dean.	2023–2024
MERT 🌐 Built a large-scale self-supervised BERT-style representation model for music audio.	2022–2023
MARBLE 🌐 Built the first unified benchmark for music understanding.	2022–2023
COIG 🌐 <i>Project Lead, COIG-PC.</i> Led the Chinese FLAN-style instruction tuning corpus effort, coordinating over one hundred contributors and converting thousands of pre-LLM-era Chinese NLP datasets into instruction format.	2022

Selected Publications and Manuscripts

Asterisks indicate equal contribution. Icons link to papers, code, or project pages.

- [arXiv 2025] YuE: Scaling Open Foundation Models for Long-Form Music Generation 📄🌐. **Ruibin Yuan**, Hanfeng Lin, Shuyue Guo, Ge Zhang, Jiahao Pan, et al.
- [CVPR 2024, highly cited] MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI 📄. Xiang Yue*, Yuansheng Ni*, Kai Zhang*, Tianyu Zheng*, et al., **Ruibin Yuan**, et al.

3. [ICLR 2024] MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training 📄. Yizhi Li*, **Ruibin Yuan***, Ge Zhang*, Yinghao Ma*, et al.
4. [NeurIPS 2023] MARBLE: Music Audio Representation Benchmark for Universal Evaluation 📄🎵. **Ruibin Yuan***, Yinghao Ma*, Yizhi Li*, Ge Zhang*, et al.
5. [ACL 2024 Main, highly cited] AnyGPT: Unified Multimodal LLM with Discrete Sequence Modeling 📄. Jun Zhan, Junqi Dai, Jiasheng Ye, Yunhua Zhou, et al., **Ruibin Yuan**, et al.
6. [COLM 2024, highly cited] RQ-RAG: Learning to Refine Queries for Retrieval Augmented Generation 📄🔍. Chi-Min Chan, Chunpu Xu, **Ruibin Yuan**, Hongyin Luo, Wei Xue, Yike Guo, Jie Fu.
7. [ICLR 2025] MuPT: A Generative Symbolic Music Pretrained Transformer 📄. Xingwei Qu, Yuelin Bai, Yinghao Ma, Ziya Zhou, Ka Man Lo, Jiaheng Liu, **Ruibin Yuan**, et al.
8. [ACL 2025 Findings] CLaMP 3: Universal Music Information Retrieval Across Unaligned Modalities and Unseen Languages 📄🔍. Shangda Wu, Zhancheng Guo, **Ruibin Yuan**, Junyan Jiang, Seungheon Doh, Gus Xia, Juhan Nam, Xiaobing Li, Feng Yu, Maosong Sun.
9. [NAACL 2025 Findings] CLaMP 2: Multimodal Music Information Retrieval Across 101 Languages Using Large Language Models 📄. Shangda Wu, Yashan Wang, **Ruibin Yuan**, Zhancheng Guo, Xu Tan, Ge Zhang, et al.
10. [ACL 2024 Findings] ChatMusician: Understanding and Generating Music Intrinsically with LLM 📄. **Ruibin Yuan**, Hanfeng Lin, Yi Wang, Zeyue Tian, Shangda Wu, Tianhao Shen, Ge Zhang, et al.
11. [arXiv 2024] CMMMU: A Chinese Massive Multi-discipline Multimodal Understanding Benchmark 📄. Ge Zhang, Xinrun Du, Bei Chen, Yiming Liang, Tongxu Luo, Tianyu Zheng, et al., **Ruibin Yuan**, et al.
12. [NAACL 2025 Findings] COIG-CQIA: Quality is All You Need for Chinese Instruction Fine-tuning 📄👥. Yuelin Bai, Xeron Du, Yiming Liang, Leo Jin, Juntong Zhou, et al., **Ruibin Yuan**, et al.
13. [arXiv 2024] OmniBench: Towards The Future of Universal Omni-Language Models 📄. Yizhi Li, Ge Zhang, Yinghao Ma, **Ruibin Yuan**, Kang Zhu, Hangyu Guo, et al.
14. [ISMIR 2023] LyricWhiz: Robust Multilingual Zero-shot Lyrics Transcription by Whispering to ChatGPT 📄. Le Zhuo, **Ruibin Yuan**, Jiahao Pan, Yinghao Ma, Yizhi Li, Ge Zhang, et al.
15. [ISMIR 2023] On the Effectiveness of Speech Self-supervised Learning for Music 📄. Yinghao Ma, **Ruibin Yuan**, Yizhi Li, Ge Zhang, Xingran Chen, Hanzhi Yin, et al.
16. [Interspeech 2022] DeID-VC: Speaker De-identification via Zero-shot Pseudo Voice Conversion 📄. **Ruibin Yuan**, Yuxuan Wu, Jacob Li, Jaxter Kim.

Industry Experience

Qwen 🌐	04/2025–Present
Research Intern, Qwen Music; Supervisor: Jin Xu 🌐	Remote
Moonshot.ai 🌐	09/2024–04/2025
Research Intern; Supervisor: Xu Tan 🌐	Remote
BAAI 🌐	2022–2023
Part-time Research Intern; Supervisor: Jie Fu 🌐	Remote
Stardust.ai 🌐	05/2020–2021
Part-time ML Consultant; Ex-engineer	Beijing, China
NetEase, Inc. 🌐	12/2019–04/2020
Intern Machine Learning Engineer	Shanghai, China
Vlahakis Recording Studio at CMU	2023
Student Audio Engineer	Pittsburgh, PA

Music Experience

Lead Guitarist and Cofounder, BUPT Slow Blues Band [Album]	2016–2021
President, BUPT Guitar Club	2017–2018